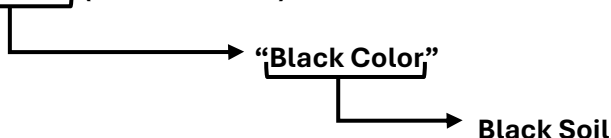


- **Introduction**

- “Chemistry” word is derived from “Chemia” (A Mishra word)



- The short name of “Chemica” is “Chemi”.
- The ancient name of Chemistry is “Chemiteching”.

- **According to Hoffmann**

- “Chemistry is the science of molecules and their transformation. It is the science not so much of one hundred elements but of the infinite variety of molecules that may be built from them”
- Chemistry also plays an important role in our daily life.
- Chemistry is also called science of “atoms and molecules”

Note:

- The black smooth soil is called “Chemi” by Egyptians.

- **Definition**

- Chemistry is the scientific study of the properties and behaviour of the matter.
OR
- Chemistry is a natural science that covers the elements that make up matter to the compounds composed of atoms, molecules and ions, their composition, structure properties, behaviour and the changes they undergo during a reaction with other substances.
OR
- Chemistry is that branch of science in which study about the composition structure, properties, behaviour and interaction of matter.
- Chemistry is also called central science because its properties occupies an intermediate position between physics and biology.
- The father of chemistry – Antonie De. Lavoisier

S.No	Contents	Page no.
1.	Chemical reaction & equation	
2.	Acid, Base and Salt	
3.	Metal and Non – metal	
4.	Carbon and it's compound	
5.	Classification of elements	

CHEMICAL REACTION & EQUATION

- Chemical Reaction

- That process in which **the substances** undergo change to produce new substances with new properties is known as chemical reactions.

For example:-

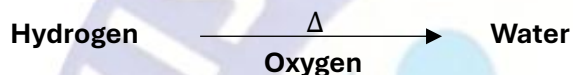
- i) When magnesium ribbon is heated, it burns in air with a dazzling white flame to form a white powder called magnesium oxide.

i.e,



- ii) When Hydrogen is burnt in air in the presence of oxygen then water is formed.

i.e,



Or,



Note:

- Souring of milk, Formation of curd from milk, Cooking of food, Digestion of food in our body, process of respiration, Rusting of Iron, burning of fuel, ripening of fruits, burning of Candel Wax, Fermentation of grapes etc. are examples of chemical reaction.

- Reactants

- Those substances which take part in a chemical reaction are called reactants.

- Products

- Those new substances with new properties are formed in a chemical reaction, called products.

For example:-



Here,

Hydrogen and Oxygen are reactants and water is the product.

Note:

1. The property of product is entirely different from the properties of **Reactants**.
2. In this Chemical Reaction first reactants are decomposed into own atoms and again they combine to form the molecule of products.
3. In a chemical equation, The reactants towards the tail while products always towards the head of arrow.

• **Characteristics of Chemical Reaction**

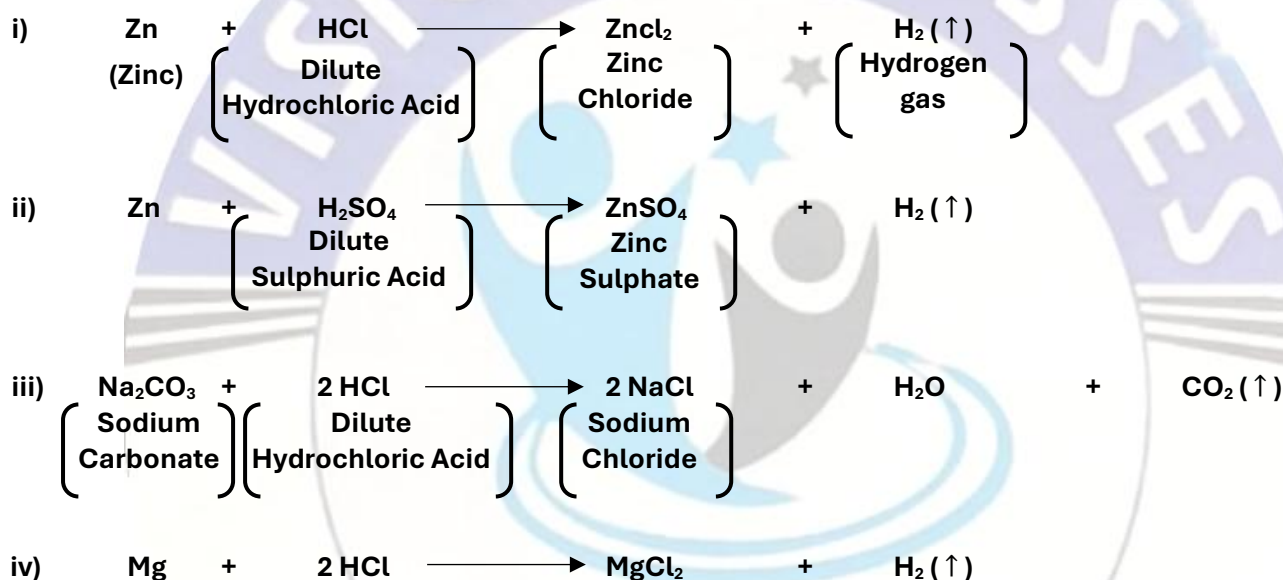
1. Evolution of gas.
2. Formation of precipitate.
3. Change in temperature.
4. Change in Color.
5. Change in State.

1. **Evolution of gas**

➤ Some Chemical reactions are characterized by the evolution of a gas.

For example:-

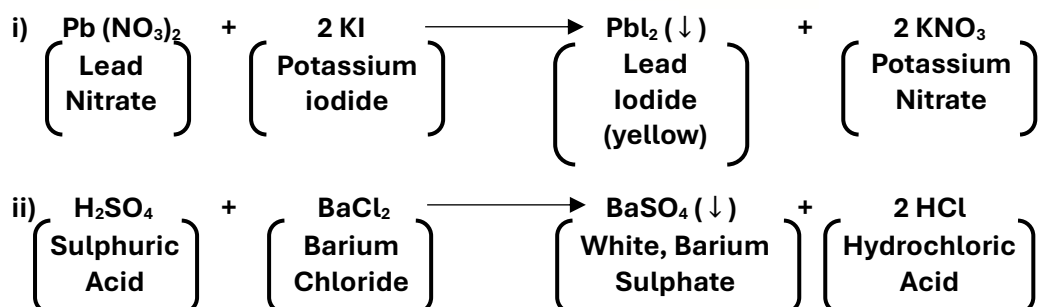
✓ Some metals like as zinc, Magnesium, iron etc. reacts with dilute acid with the evolution of Hydrogen gas-

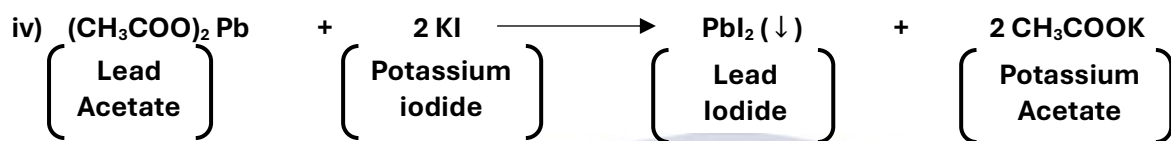
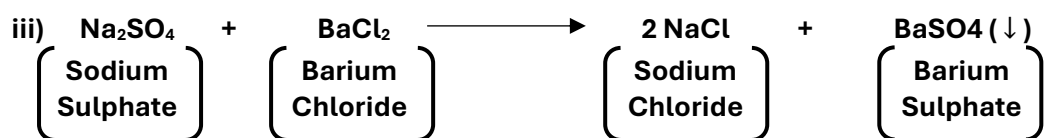


2. **Formation of precipitate**

- A precipitate is a "Solid Product" which separates out from the solution during a chemical reaction and this type of reactions are called precipitation reactions.
- A precipitate can be formed by mixing aqueous solutions of reactions when one of the products is insoluble in water.
- Generally, the color of precipitate is white but in some cases it is yellow.
- Some chemical reactions are characterized by the formation of a precipitate.

For example:-

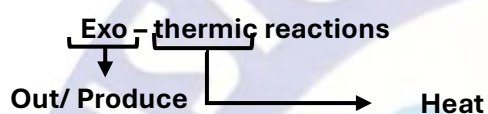




3. Change in Temperature

- In some chemical reactions are characterized by a change in temperature.

Case 1:

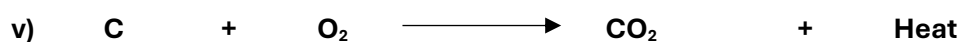
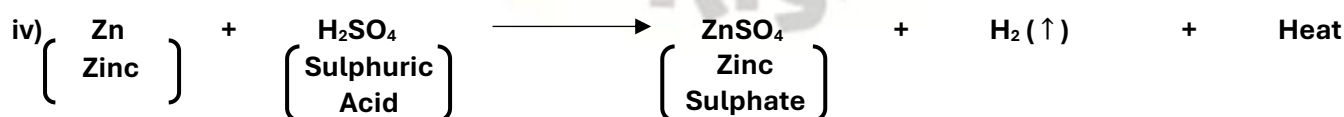


- Those reactions in which heat is produced during the chemical reaction are called exo - thermic reactions.

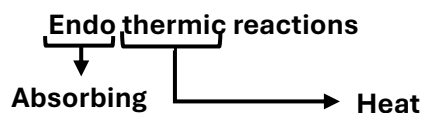
For example:-



Or,



Case 2:



- Those chemical reactions in which heat is absorbed to occur the chemical reaction are called endothermic reactions.

For example:-



Or,



4. Change in Color

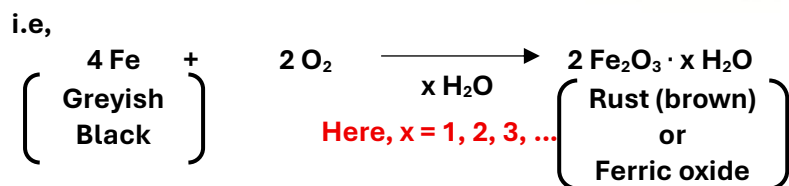
- Some Chemical reactions are characterized by a change in color.

For example:-

- i) The yellowish colored chlorine is passed through colorless potassium iodide then formed potassium chloride and brown colored iodine.



- ii) When iron (greyish black) is left exposed to air for a long time, the outer surface of iron gets covered with brown coating of rust.



- iii) When an iron nail is kept in copper sulphate solution then, the colour of solution converted into a light green because iron displaced copper from copper sulphate.



v) The Chemical reaction between sulphuric dioxide gas (colorless) and acidified potassium dichromate (orange) solution is characterized by a change in color from orange to green.

➤ **Some chemical reactions are characterized by a change in State.**

✓ When a wax candle is burned in air then it melted in liquid and produced carbon dioxide gas where wax vapor is liquid while CO_2 is gas.

➤ The process of representing a chemical reaction with the help of symbols and formulae of the substances involved in it is known as Chemical equation.

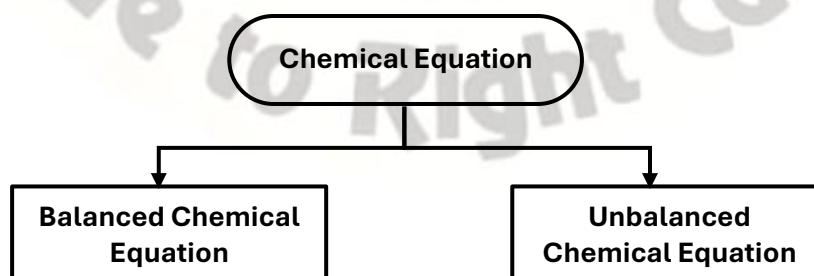
i) $\text{H}_2 + \text{Cl}_2 \longrightarrow 2\text{HCl}$

↓
Reactants

↓
Product

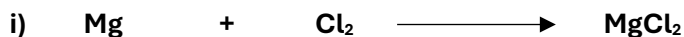


- A Chemical equation is a Short – hand method of representing a Chemical reaction.



➤ That chemical equations in which an equal number of atoms of different elements on both sides of the equation is called balanced chemical equations.

For example:-



	Left (Reactant)	Right (Product)
No. of Mg atoms	1	1
No. of Cl atoms	2	2

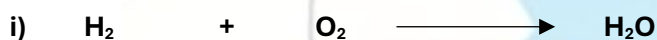


	Left (Reactant)	Right (Product)
No. of Fe atoms	1	1
No. of H atoms	2	2
No. of S atoms	1	1
No. of O atoms	4	4

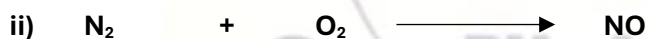
2. Unbalanced Chemical Equation

- That Chemical equation in which an unequal number of atoms of different elements on both sides of the equation is called Unbalanced Chemical equation.

For example:-



	Left (Reactant)	Right (Product)
No. of H atoms	2	2
No. of O atoms	2	1



	Left (Reactant)	Right (Product)
No. of N atoms	2	1
No. of O atoms	2	1

Note:

- A balanced chemical equation has equal masses of various elements in reactants and products.
- An Unbalanced equation has unequal masses of various elements in reactants & products.

▪ **The Rule of Writing of balanced Chemical Equations**

Rule 1:

- First we write the reactants in their symbols and formulae on the left hand side similarly products are placed on right side of the chemical equation.

Rule 2:

- A plus (+) sign is added between the formulae of the reactants and products of the chemical equation.

Rule 3:

- An arrow (→) sign is **inserted** between the reactants and products to show which the reaction is occurring



In this chemical equation, A & B are called the reactants while C & D are called products, the arrow indicates the reaction proceeds towards the formation of C and D.

▪ **The Chemical equation can be made more informative in three ways.**

- By indicating the “physical State” of the reactants and products.
- By indicating the “heat Changes” taking place in the reaction.
- By indicating the “Conditions” under which the reaction takes place.

▪ **Balancing the Chemical Equation**

- The process of making the number of different types of atoms equal on both the sides of an equation is called balancing of equation.

Note:

- The Chemical equations are balanced to satisfy “the law of conservation of mass in the chemical reaction”.

According to the law of conservation of matter/mass, “Mass can neither be created nor be destroyed in a Chemical reaction. It can only be transformed from one substance to other substance.

▪ **The Balancing of Chemical equation can be done by two ways/methods**

- Hit and trial method (for simple equation).
- Partial equation method (for complex equation).
- a, b, c & d method.

• **In Balancing the Chemical equation**

- The correct symbols and formulae for each substance involved in the reaction must be known.
- None of the subscripts of the formulae on the left-hand side can be change in the balancing in the equation.

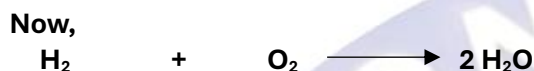
i) **Hit and Trial Method**

➤ This method is only applicable for balancing simple chemical equations.

For example:-

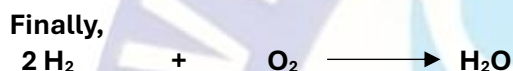


	Left Side	Right Side
No. of H - atoms	2	2
No. of O - atoms	2	2×1

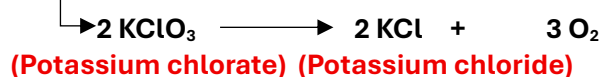
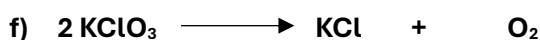
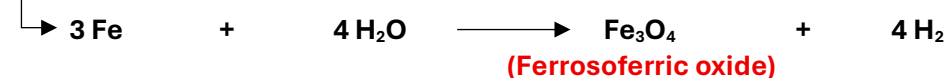
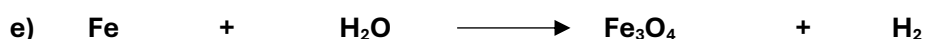
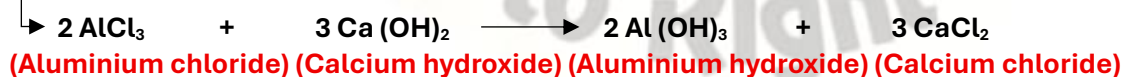
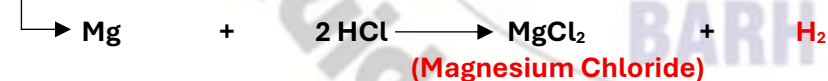
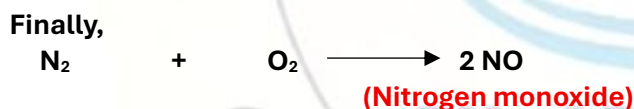


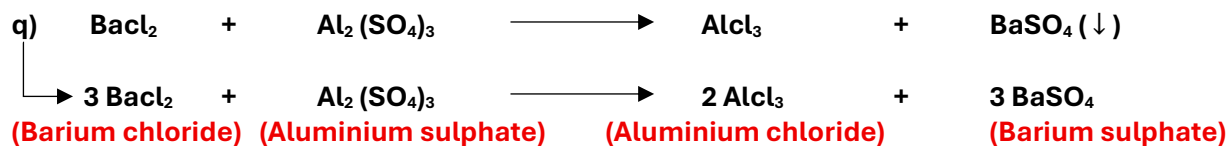
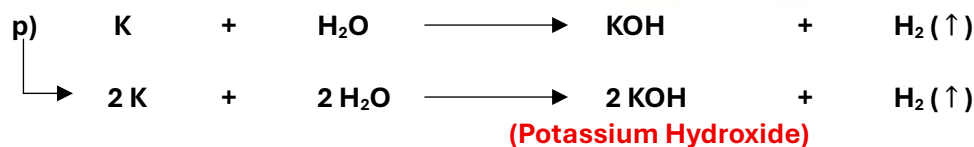
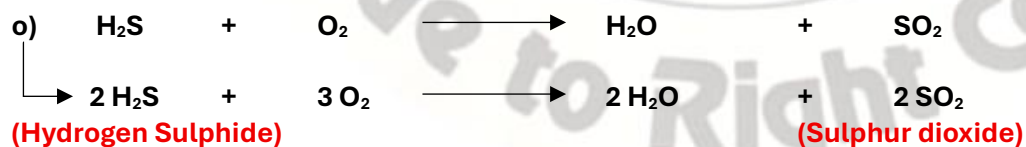
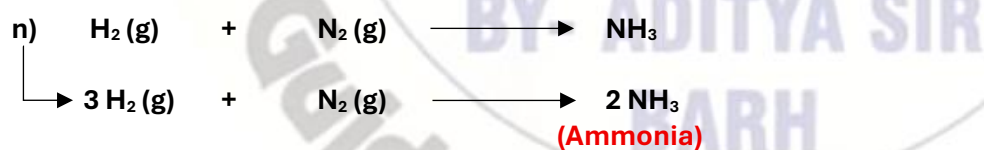
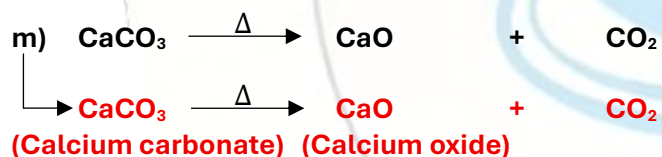
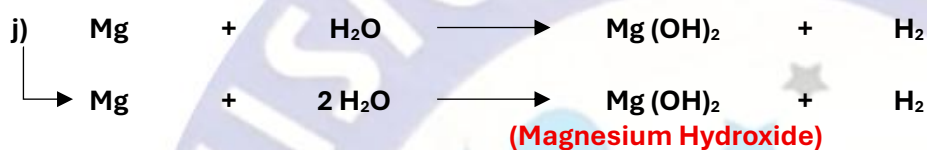
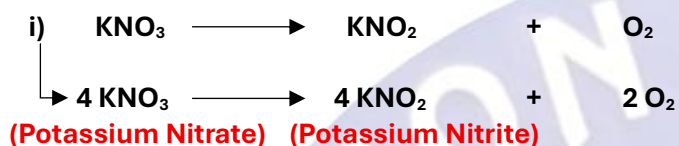
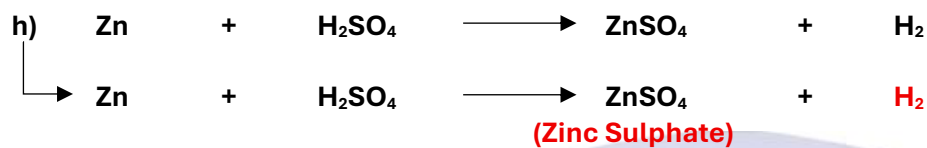
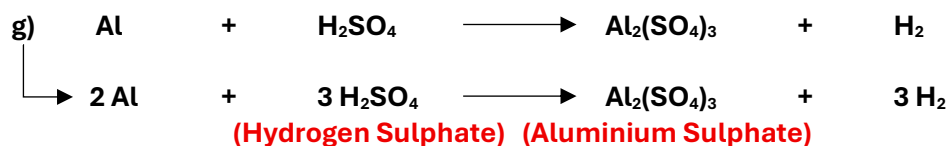
Again,

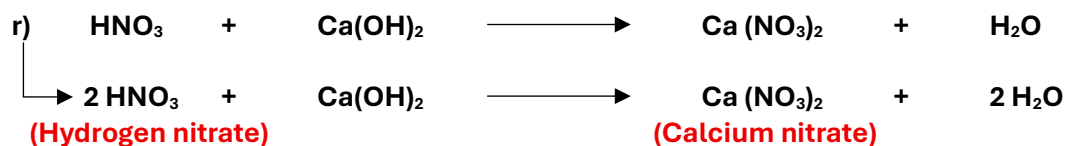
	Reactant (Left)	Product (Right)
No. of H - atoms	2×2	4
No. of O - atoms	2	2



	Reactant (Left)	Product (Right)
No. of N - atoms	2	2×1
No. of O - atoms	2	2×1







ii) Balancing by a, b, c & d ... method

For example:-



$$\begin{aligned} a &= e && \text{----- (i)} \\ 3a &= 2e + f && \text{----- (ii)} \\ b &= d && \text{----- (iii)} \\ 2b + c &= 4d && \text{----- (iv)} \\ 2c &= 2d + f && \text{----- (v)} \end{aligned}$$

Let,

$$a = 1$$

$$\therefore a = e = 1 \quad [\text{from equation (i)}]$$

$$\Rightarrow 3 \times 1 = 2 \times 1 + f \quad [\text{from equation (ii)}]$$

$$\Rightarrow f = 3 - 2 = 1$$

[from equation (v)]

$$\Rightarrow 2c = 2d + 1$$

$$\Rightarrow 2c - 2d = 1 \quad \text{----- (vi)}$$

[from equation (iii) & (iv)]

$$\Rightarrow 2d + c = 4d$$

$$\Rightarrow c - 2d = 0 \quad \text{----- (vii)}$$

[Subtract equation (vii) from equation (vi)]

$$2c - 2d = 1$$

$$c - 2d = 0$$

$$\begin{array}{r} - \\ + \\ - \end{array}$$

$$c = 1$$

[Putting the value of c in equation (vii)]

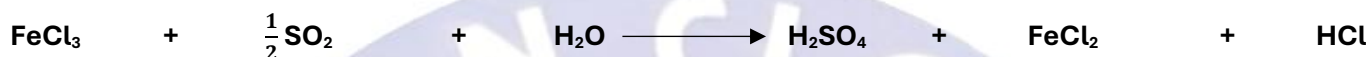
$$\Rightarrow 1 - 2d = 0$$

$$\Rightarrow 1 = 2d$$

$$\Rightarrow d = \frac{1}{2}$$

[from equation (iii) $b = d = \frac{1}{2}$]

Finally,



- **Advantage of writing of chemical equation**

- The representation of chemical reaction easy. It save time and space for writing.
- In order to prepare a definite amount of product the amount of reactant to be used can be calculated accurately.
- The same chemical symbols or for Mulae are used all over the world **due** to which the scientists do not face difficulties in understanding chemical equation.

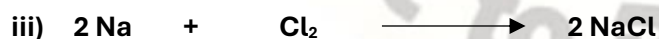
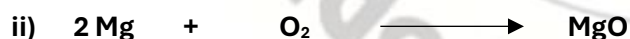
CLASSIFICATION OF CHEMICAL EQUATION

1. Combination Chemical reactions

- Those chemical reactions in which two or more substances combine to produce new substance are called combination Chemical reactions.
- These are also called synthesis chemical reactions.
- These Chemical reactions are represented by $A + B \longrightarrow AB$.

For example:-

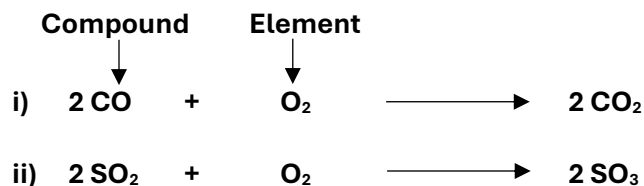
- **Case 1: The Combination of elements**



- **Case 2: The Combination of compound and compound**



▪ **Case 3: The Combination of Compound and element**



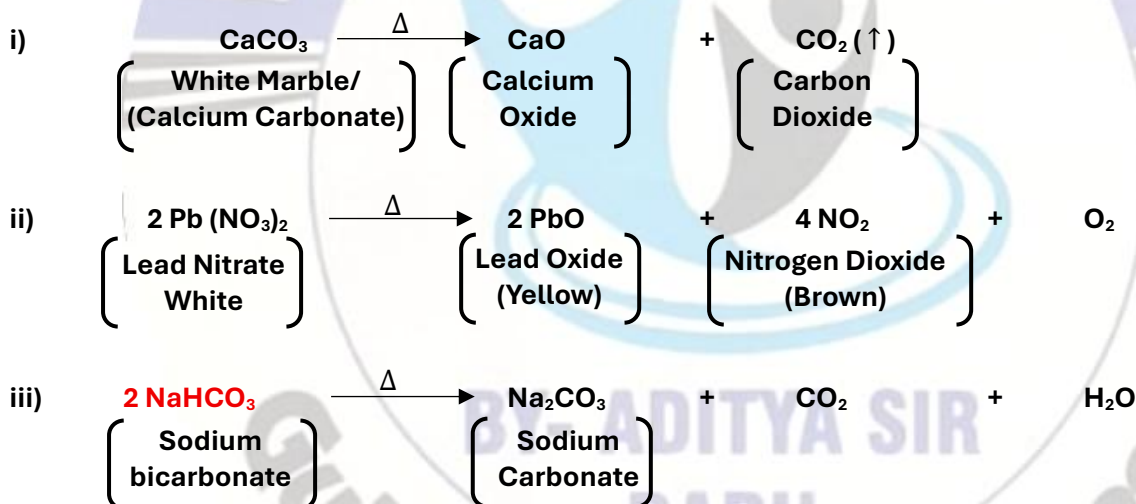
2. Decomposition reaction

- Those Chemical reactions in which a compound is broken/decomposed into two or more simpler substances are called decomposition chemical reactions.
- These are represented by $AB \longrightarrow A + B$

Note:

Decomposition $\neq$ Composition

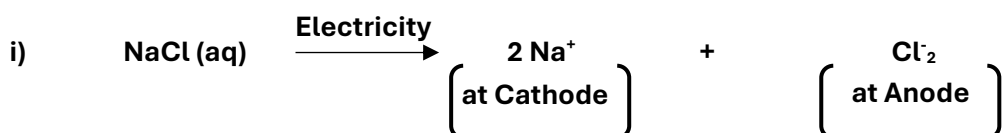
For example:-

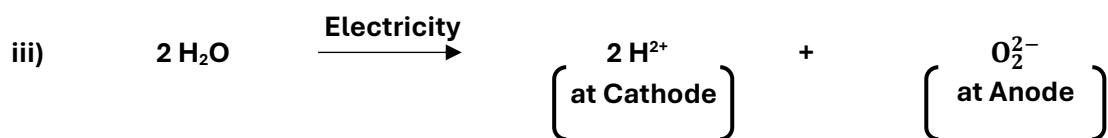
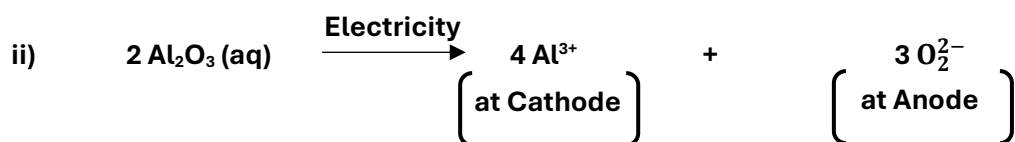


3. Electrolytic decomposition reactions

- Those chemical reactions in which like as metal oxide and metal chloride are decomposed by passing electricity through them in their molten state these type of reactions are called electrolytic decomposition reaction.
- In this reaction, metal are decomposed on cathode (–) and oxygen/chlorine gas is decomposed on anode (+).

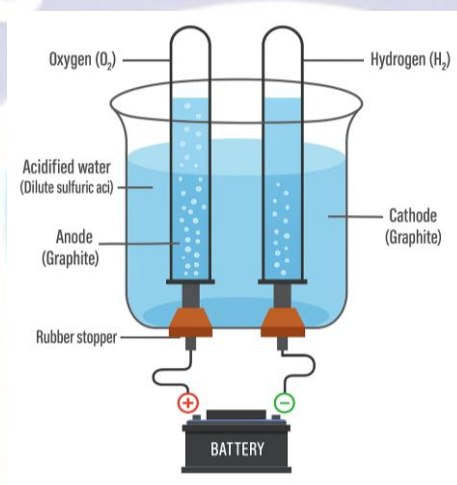
For example:-





Note:

Electrolysis of water

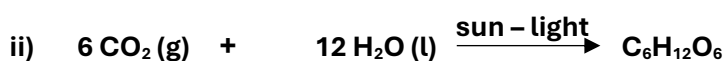
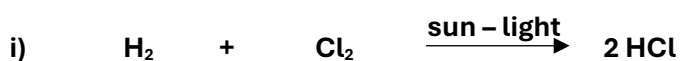


- 1) Electrolytic decompositions are used to obtain metal from metal oxide or metal chloride.
- 2) Digestion of food in our body is a decomposition reaction because the food we eat contains carbohydrates and proteins carbohydrates decomposes into glucose.

4. Photo Chemical reaction

- Those chemical reactions which take place in the presence of the sun light. These reaction are called photo – chemical reactions.

For example:-





5. Single displacement reactions

- Those reaction in which an atom or group of atoms present in a molecule are displaced by another atom, are called single displacement reaction.

▪ Case 1: Displacement of less reactive metal by more reactive metal.

For example:-



Here,

Fe displaced Cu because Fe is more reactive than Cu.



Cu is more reactive than Ag



▪ Case 2: Displacement of less reactive non – metal from its compound by more reactive non – metal.

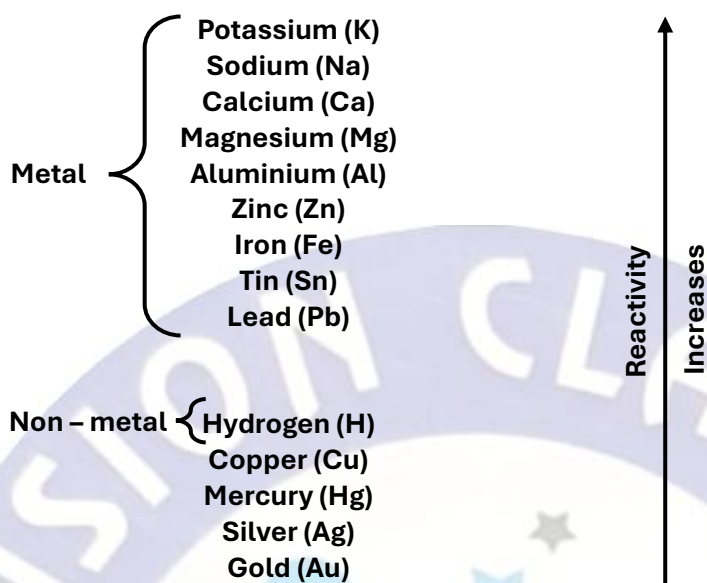
For example:-



▪ Case 3: Displacement of hydrogen from its compound by some reactive metal.



▪ Reactivity series of metal



Note:

- Reactivity of metals depend on their donating capacity of electrons.
i.e,
Reactivity of metal $\propto$ Donating capacity of electron
- Higher the donating capacity higher the reactivity of metal.

▪ Reactivity series of non – metal

Fluorine (F)
Chlorine (Cl)
Bromine (Br)
Iodine (I)
Astatine (At)

Reactivity
Increases

Note:

- Reactivity of non – metals depends on gaining capacity of electrons.
i.e,
Reactivity of non – metals $\propto$ Gaining capacity of electron
- Higher the gaining capacity higher the reactivity of non – metal.

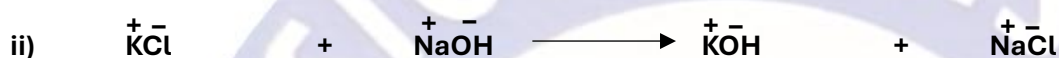
6. Double displacement reaction

- Those reactions in which only two compounds exchange their ions to form two new compounds. These reactions are also called double displacement reactions.
- These can be represented by



- These reactions are also called metathesis reactions.

For example:-



7. Precipitation reaction

- Those chemical reactions in which solid substance get separated from the solution. These reactions are called precipitation reactions.

- **Precipitate**
- The separated solid from the solution is called precipitate.

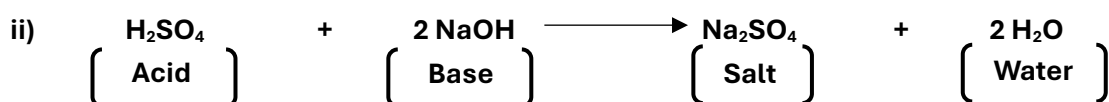
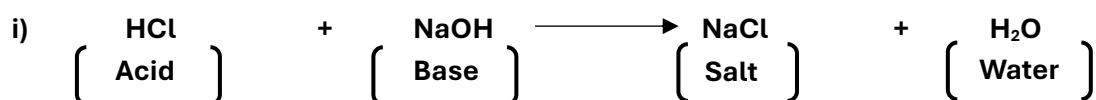
For example:-



8. Neutralisation reaction

- Those chemical reaction in which an acid strongly reacts with a base to form neutral salt and water, called neutralization reaction.

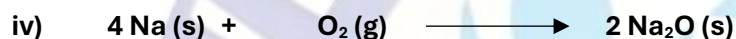
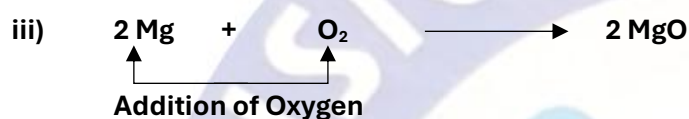
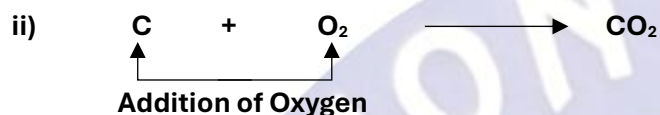
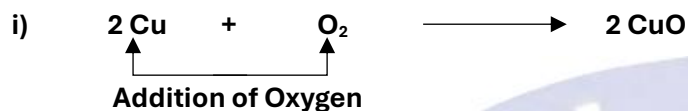
For example:-



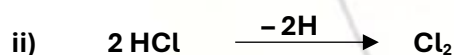
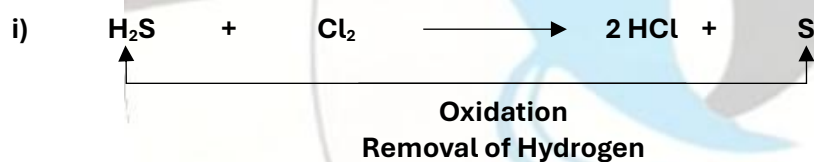
9. Oxidation chemical reaction

- Those chemical reactions in which either addition of oxygen to an element/compound or removal of hydrogen from a compound, called Oxidation Chemical reactions.

▪ **Case 1: Addition of Oxygen**



▪ **Case 2: Removal of Hydrogen**



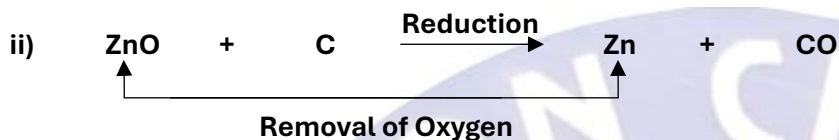
Note:

- Respiration, digestion of food, rancidity of food, Corrosion of metal, Combustion etc. are example of oxidation Chemical reactions.

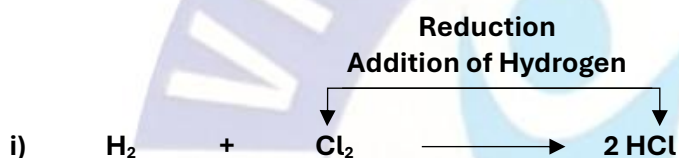
10. Reduction Chemical reaction

- These Chemical reactions in which either removal of oxygen from a compound or addition of hydrogen to an element/compound, called reduction chemical reaction.

- **Case 1: Removal of Oxygen from a compound**

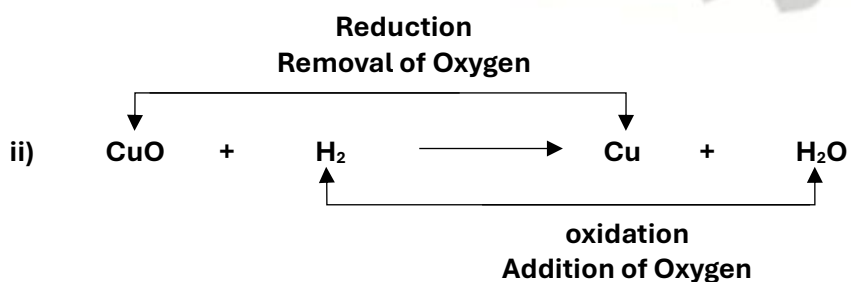
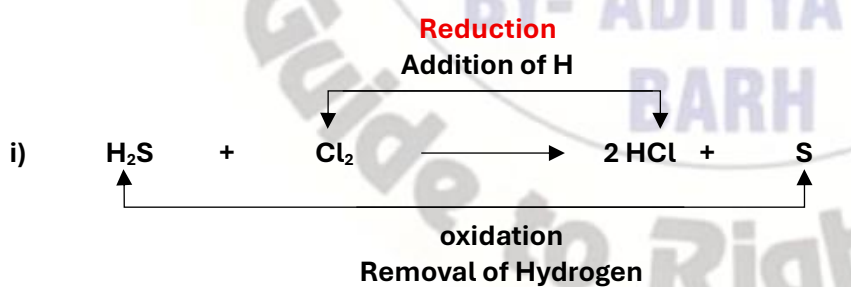


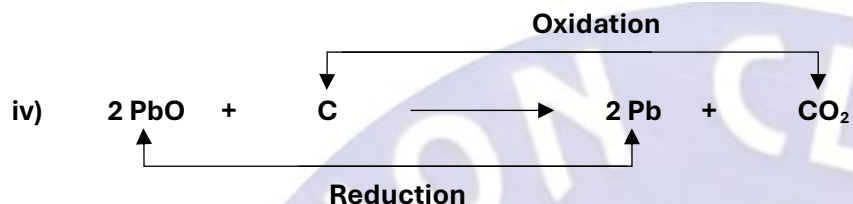
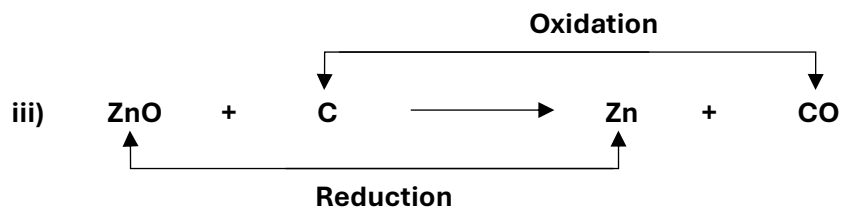
For example:-



➤ The Oxidation and reduction both always **takes place** simultaneously. These type of reaction are called redox reaction.

For example:-





- **Oxidising Agent**

- When Oxygen is removed from any substance (element/compound) then we said the compound is reduced this reduced substance is called oxidising agent.

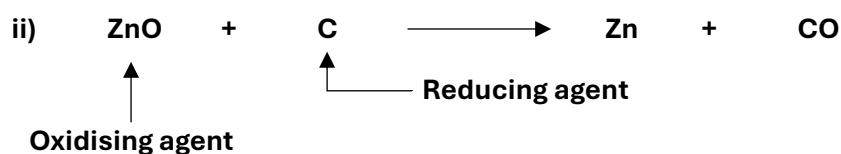
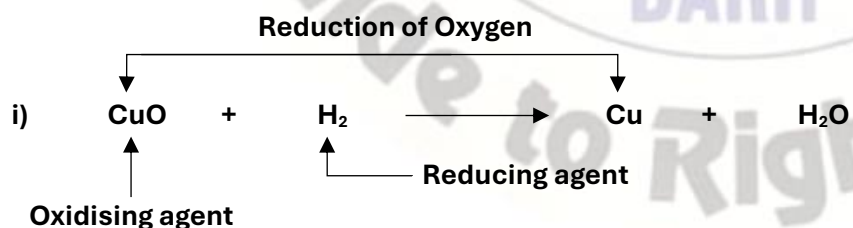
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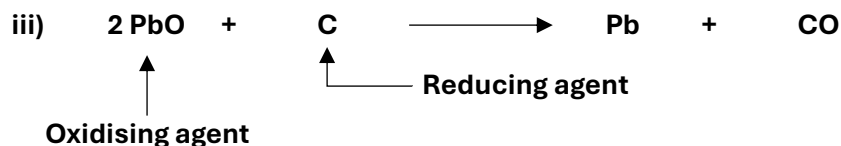
- Those substances which oxidised element/compound are called oxidising agent.
- An Oxidising agent oxidised something else.

- **Definition of Oxidising agent in the term of electron.**

- An Oxidising agent must gain electrons by another substance.
- Oxidising agent having capacity (nature) to accept electrons due to which it is also called electrons acceptor.
- Thus, all non – metals are called oxidising agent.

For example:-





Note:

1. All 'ic' state of metal compounds are oxidising agent.

For example:-

✓ Ferric Oxide, Plumbic Oxide, Cupric Oxide etc.

2. The Strongest Oxidising agent is fluorine, because fluorine is high reactive non-metal.

3. The weakest Oxidising agent is Lithium.

Name of Oxidising agent	Molecular formula
Ozone	O_3
Potassium permanganate	KMnO_4
Potassium dichromate	$\text{K}_2\text{Cr}_2\text{O}_7$
Manganese di-oxide	MnO_2
Nitric acid	HNO_3
Sulphuric acid	H_2SO_4
Ferric chloride	FeCl_3
Plumbic chloride	PbCl_4
Fluoride	F_2
Chloride	Cl_2
Iodine	I_2
Nitrogen	N_2
Hydrogen peroxide	H_2O_2
Ferric sulphate	$\text{Fe}_2(\text{SO}_4)_3$

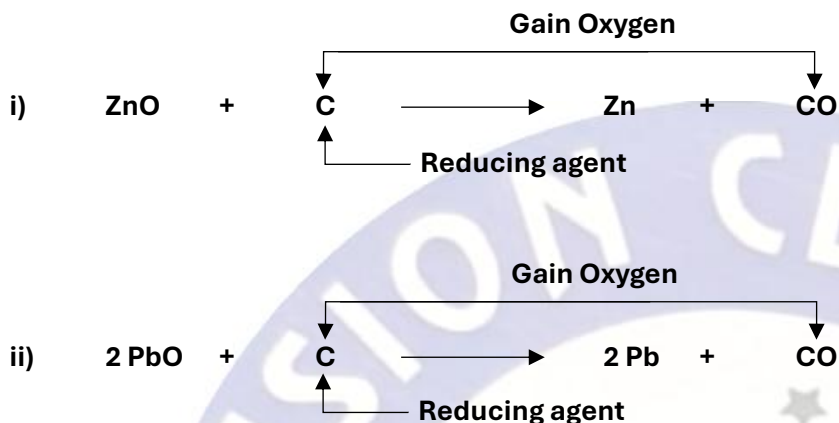
• **Reducing agent**

- When any substance gain the oxygen then we said that the substance is oxidised and these oxidised substances is called reducing agent.
- A reducing agent is reduced other substances.

▪ **Definition of reducing agent in the form of electron**

- A reducing agent must loss e^- (s).
- It is also called 'reductant' or 'reducer'.

For example:-



Note:

1. All 'us' state of metal compounds are called reducing agent and all metals are called reducing agent.
2. The Strongest reducing agent is Lithium.
3. The weakest reducing agent is Hydrogen.

✓ **Another examples of reducing agents**

Name of reducing agent	Molecular Formula
Hydrogen Sulphide	H_2S
Hydrogen Iodine	HI
Potassium Iodine	KI
Ferrous Sulphate	FeSO_4
Stannous Chloride	SnCl_2
Plumbus Chloride	PbCl_2
Lithium	Li
Sodium	Na
Potassium	K
Magnesium	Mg
Calcium	Ca

Special Note:

- Some substances which behaves as oxidising agent and reducing agent.

For example:-

- ✓ Iodine (I_2), Nitrous Acid (HNO_2), Sulphur dioxide (SO_2), Hydrogen per oxide (H_2O_2), Ozon (O_3) etc.

• **Defining of Oxidation Chemical reaction and reduction Chemical reaction in the term of electron**

▪ **Oxidation**

➤ Oxidation is either loss of electron or increases in oxidation state by a molecules, atoms, ions etc.

▪ **Reduction**

➤ Reduction is either the gain of electrons or decrease in oxidation state by a molecules, atoms, ions etc.

▪ **Oxidation number**

➤ The Oxidation number is defined as the formal charge on the atom appears to have when electrons are counted.

➤ It is either positive or negative.

For example:-

	Oxidation No.
Na ⁺	+1
Cl ⁻	-1
O ⁻²	-2
N ⁻³	-3
Al ⁺³	3

Q. Find the Oxidation number of

a) MgCl₂

Ans Let Oxidation number of Mg = x

$$\Rightarrow x \times (-2) = 0.$$

$$\Rightarrow x = 2.$$

b) NO₃

Ans $x = -3 \times (-2) = 6$

a) MgCl₂ → Formula Charge = 0 (Neutral)

$$\Rightarrow 2 \times (-1) \times 2 = 0$$

$$\Rightarrow \text{O.N} = 0$$

c) SO₄⁻²

$$\Rightarrow 6 + (-2) \times 4$$

$$\Rightarrow 6 - 8 = -2$$

$$\Rightarrow \text{O.N of SO}_4 = -2$$

b) NO₃⁻¹

$$\Rightarrow 5 + (-2) \times 3 = 5 - 6 = -1$$

$$\Rightarrow \text{O.N} = -1$$

d) Fe₂O₃

$$\Rightarrow 3 \times 2 + (-2) \times 3$$

$$\Rightarrow 6 - 6$$

$$\Rightarrow 0$$

Note:

➤ The Oxidation **no.** of a compound is the formal **charge** on the compound.

Q. Find the Oxidation no. of Sulphur (S) in $\text{H}_2\text{S}_2\text{O}_7$.

Ans Let,

Oxidation no. of S = x .

In $\text{H}_2\text{S}_2\text{O}_7$

$$\Rightarrow 1 \times 2 + x \times 2 + (-2) \times 7 = 0$$

$$\Rightarrow 2 + 2x - 14 = 0$$

$$\Rightarrow 2x = 12$$

$$\Rightarrow x = 6$$

$$\therefore x = 6$$

$\therefore$ O.N of Sulphur in $\text{H}_2\text{S}_2\text{O}_7$ is 6.

Q. Find the Oxidation no. of Oxygen in Fe_2O_3 .

Ans Let,

Oxidation no. of O = x .

In Fe_2O_3

$$\Rightarrow 3 \times 2 + x \times 3 = 0$$

$$\Rightarrow 6 + 3x = 0$$

$$\Rightarrow 3x = -6$$

$$\Rightarrow x = -2$$

$$\therefore x = -2$$

$\therefore$ O.N of Oxygen in Fe_2O_3 is -2.

• **Combustion/Burning**

- That process in which any combustible substance is burning in the presence of Oxygen, called Combustion.
- In this process of combustion heat and light are produce.
- The combustion reaction may be considered as a fast oxidation reaction.

For example:-



Combustion is not possible in the absence of Oxygen, nitrogen and Chlorine.

- The supporters of combustion are oxygen, nitrogen and chlorine.
- Carbon dioxide (CO_2) is non supporters of combustion.

▪ **Condition necessary for combustion**

- i) Presence of combustible substance.
- ii) Presence of supporters substances of combustible substance oxygen, nitrogen and chlorine.
- iii) To obtain ignition temperature.

- **Ignition temperature**

- The minimum temperature at which any substance starts to burn is known as ignition temperature.

I.G of **Coal** > Dung Cake > Paper > Petrol > L.P.G

- **Combustible substances**

- Those substance which burn in the process of **combustion** substances.

For example:-

- ✓ Paper, wood, Coal, Petrol, Diesel etc.

- **Non – Combustible substances**

- The substances which do not burn in the process of combustion is called Non - combustible substances.

For example:-

- ✓ Iron, Stone, Bricks, sand etc.

Special Note:

- Combustion is also occurred in absence of oxygen.

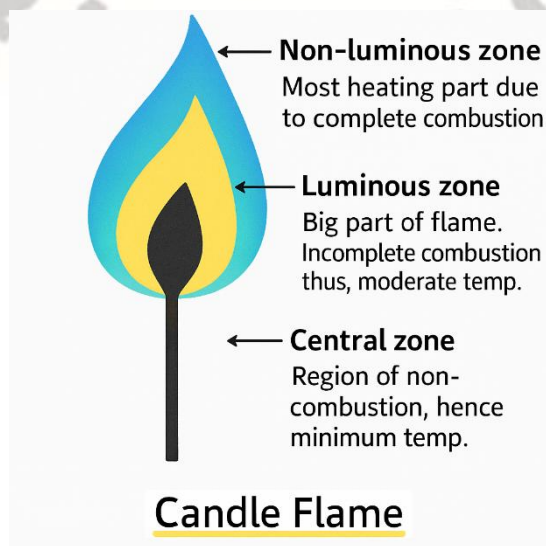
For example:-

- The metal magnesium from magnesium nitrite by burning in nitrogen.
- When any burning candle is kept in a container which is filled with chlorine then it keeps burning and forms black smoke.



- **Flame**

- When any substance is burning in the presence of oxygen then we said that flame is produced.



Types of Flame

1. Luminous Flame.
2. Non – Luminous Flame.

1. Luminous Flame

- That flame which produce with the evolution of heat and light is called luminous flame.
- It is yellow in color because of incomplete combustion **due to insufficient amount of oxygen**.

For example:-

- ✓ Kerosene lamp

2. Non – Luminous Flame

- That flame which produce with the evolution heat and very little light is called non – luminous flame.
- It is blue in color because complete combustion.

For example:-

- ✓ Kerosene, Stove, L.P.G gas, Burner etc.

• L.P.G (Liquified Petroleum gas)

- Petroleum gas is a combination of Butane (C_4H_{10}), Propane (C_3H_8) and Ethane (C_2H_6).
- The main component of L.P.G butane (C_4H_{10}).
- Petroleum gas is best fuels because it burns fastly and give enough heat.
- It becomes liquid when pressure increased and kept in cylinder used to cook food.



1 gram L.P.G produce 50 KJ of heat after burning.

The L.P.G cylinder a bad smellier gas Ethyl mercaptor (C_2H_5SH) which utters us when gas leaks.

▪ Effects of Oxidation and reduction in normal life

1. Rancidity of food

- When oil and fats containing substances are left for a long time, they undergo oxidation and become rancid. As a result, their smell and taste become unpleasant. This property of oil and fats substances is called rancidity.

Note:

- Oil and fats shows rancidity due to oxidation.

▪ The rancidity of food can be prevented by following ways:

- i) Same substances called anti-oxidants may be added to prevent oxidation of food.

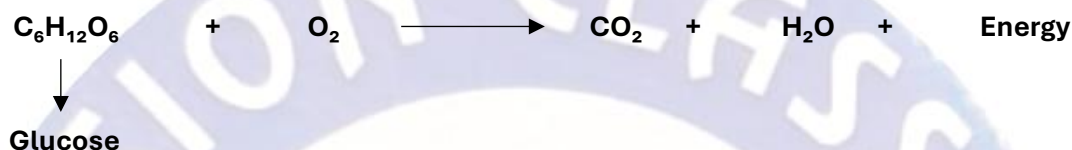
For example:-

- ✓ BHA (Butylated Hydroxyanisole), BHT (Butylated Hydroxytoluene).

- ii) Food materials may be kept at very low temperature (like as in refrigerator) to protect food from being oxidized.
- iii) Food may be preserved in airtight containers.
- iv) Keep food substances away from light.
- v) Packing fats and oils containing foods filled with nitrogen gas.

2. Digestion of Food

- Our food like rice, potatoes, bread etc. mainly contains carbohydrates. In the process of digestion, this starch is broken down into glucose which combines with oxygen and gets transformed through respiration in our cells and changes into carbon dioxide and energy. This carbon dioxide is released from our body when we breathe out, and energy maintains temperature in our body and helps to do metabolic activities.



3. Corrosion

- That process in which metals are eaten up gradually by the action of air, moisture, or a chemical on their surface is called corrosion.

For example:-

- ✓ Rusting of Iron

A layer of iron on cutting fruits. (“Don’t eat directly because it is harmful for health.”)

Explanation:

- The human body requires ferrous oxide (FeO) to form blood, but a layer of iron on cutting fruits contains ferric oxide (Fe₂O₃) which is harmful to us.

N.C.E.R.T. Question Answers

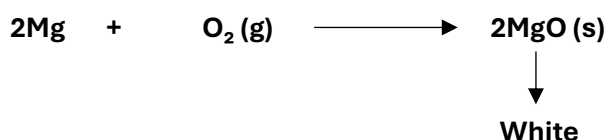
Q1. Why does a burning candle give out light?

Ans A burning candle gives out light, i.e., it is luminous.

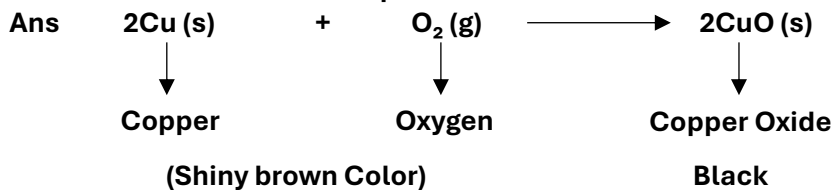
This is chiefly because solid particles of carbon are present in the flame and they become incandescent when the candle burns.

Q2. Why should magnesium ribbon be cleaned before burning in air?

Ans Since magnesium is a highly reactive metal, its surface is tarnished due to the formation of a thin layer of basic magnesium **oxide** with atmospheric oxygen, which appears as combustion. Hence, magnesium ribbon is cleaned before burning with sandpaper.



Q3. A shiny brown coloured element 'X' on heating in air becomes black in colour. Name the element 'X' and the black coloured compound formed.



i.e, **X** = Copper
& Black colored substance = CuO

Q4. Why do we apply paint on iron objects?

Ans When iron is left in the atmosphere, then iron reacts with oxygen and moisture. Due to this reaction, a brown layer is formed on it.

This brown layer is known as rust. So we apply paint on iron articles to protect them from rusting and also to decorate them.



Q5. Oil and **Fats** Containing food items are flushed with nitrogen. Why?

Ans When oils and fat containing foods are left for a long time, they undergo oxidation and become rancid. So, they are flushed with nitrogen in order to prevent them from getting oxidized.

Q6. Why is respiration considered as exothermic reaction? Explain.

Ans We get energy from food when we eat. During digestion, the carbohydrates of the food are broken into glucose. This glucose combines with the oxygen in our body and provides us energy to do work and stay alive.